PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Elizabeth Ogijoh
Program/Batch	B084-Networkwalks
Date	11 November 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (written permission already secured) 2. My own local lab network (10.0.2.0/24)
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: In Progress

1. Liability Disclaimer

I carried out these activities only on systems and devices where I had written permission, or that I own myself. Everything here is for education and research. Please do not use any of it to break the law. The instructor, the authors and Networkwalks are not responsible for what anyone does with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers two practical modules from Week 2 of my cybersecurity program at Networkwalks. In W2-PM1 I footprinted the networkwalks.com domain with six Kali Linux tools. In W2-PM5 I scanned my own local network with Zenmap. The first module is about collecting public information and the second is about finding out which devices are alive on a network, so together they show the early steps an attacker (or a defender thinking like one) would take.

The footprinting commands were run from Kali Linux on 9 November 2026 and the Zenmap scan was run on a Windows PC on 10 November 2026. For each step I list the command, what I saw, and why it matters. The full terminal output and screenshots are in Section 8.

3. Tools Used

These are the tools used in this report and what each one was for.

Tool	Purpose
Kali Linux & Windows	Operating systems used for the reconnaissance and scanning work.
WHOIS	Domain registration details (registrar, dates, name servers).
whatweb	Fingerprints web technologies (server, CMS, plugins, IP).
nslookup	Resolves the domain name to its IP address using DNS.
curl -I	Reads the HTTP response headers of the website.
wafw00f	Detects whether a Web Application Firewall protects the site.
dnsrecon	Enumerates DNS records (SOA, NS, MX, A, TXT/SPF, SRV).
Zenmap (Nmap GUI)	Ping scan of the local subnet to find live hosts, IPs and MAC addresses, plus the topology view.
Windows CMD	Used to check the local IP address and MAC address of my PC.

4. Activities Performed

4.1 Footprinting & Reconnaissance (W2-PM1)

The target for this part was `networkwalks.com`. I ran six tools, and each one gave me a different slice of information about the same domain.

Task 1: WHOIS, domain registration

Command: `whois networkwalks.com`

- The domain is registered through **GoDaddy.com, LLC** (IANA ID 146).
- It was created on **6 November 2019**, last updated on 12 November 2025 and expires on **6 November 2027**.
- The name servers are **NS6135.HOSTGATOR.COM** and **NS6136.HOSTGATOR.COM**, so the site is hosted at HostGator.
- Four lock statuses are set (delete, renew, transfer and update prohibited) and DNSSEC is shown as unsigned.

Why it matters: the registrar, hosting provider and key dates tell an attacker who to target for social engineering or a hijack attempt. The locks are good news because they make an unauthorised transfer harder.

Task 2: WhatWeb, web technologies

Command: `whatweb networkwalks.com`

- The http:// address returns a **301 redirect** to https://, and the HTTPS page answers 200 OK.
- The site runs on **Apache** with **WordPress 7.1.4** and the **WordPress Download Manager 3.3.76** plugin.
- It also uses jQuery 3.7.1, Bootstrap and Google Tag Manager, and the page title is "Networkwalks Academy".
- The IP is 192.232.216.135, and the contact email **info@networkwalks.com** appears in the page.

- The `__wpdm_client` cookie is set with the HttpOnly flag.

Why it matters: knowing the CMS, plugin and exact versions is a big head start, because an attacker can search for issues that affect those specific versions. A published email address can also be used for phishing.

Task 3: Nslookup, domain to IP

Command: `nslookup networkwalks.com`

- Using Google DNS (8.8.8.8), the domain resolved to **192.232.216.135**. The answer was non-authoritative, which just means it came from a resolver and not from the domain's own name server.

Why it matters: the IP is the starting point for any later port scan or service check against the host, and it matches what WhatWeb and DNSRecon reported.

Task 4: Curl -I, HTTP response headers

Command: `curl -I https://networkwalks.com`

- The server answered **HTTP/2 200** and identified itself as **Apache**.
- The `link` header exposes the WordPress REST API at **/wp-json/** and shows the home page is WordPress page ID 53.
- Cache headers (`x-nginx-cache: WordPress`, `x-endurance-cache-level: 0`) show the caching layer in front of the site.
- The cookie carries the `secure` and `HttpOnly` flags, and a `referrer-policy` and `permissions-policy` are present.

Why it matters: headers are free information. Here they confirm WordPress and point to the REST API, which is a common place to continue enumeration. On the positive side, the security-related headers and cookie flags show some hardening is in place.

Task 5: Wafw00f, firewall detection

Command: `wafw00f networkwalks.com`

- Wafw00f v2.4.2 used only 2 requests and reported the site is behind **ModSecurity (SpiderLabs)**.

Why it matters: it is good that a WAF is there, but the fact that it can be identified so quickly tells an attacker which rule set they would need to work around.

Task 6: DNSRecon, DNS records

Command: `dnsrecon -d networkwalks.com`

- **SOA / NS:** ns6135.hostgator.com (50.87.144.87) and ns6136.hostgator.com (192.232.216.131).
- **BIND version:** both name servers revealed **"9.16.23-RH"**.
- **MX:** mail.networkwalks.com on 192.232.216.135, the same IP as the website.
- **A:** networkwalks.com resolves to 192.232.216.135.
- **TXT:** an SPF record (`v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all`) and a Google site verification token.
- **SRV:** eight `_autodiscover._tcp` records pointing to cPanel email discovery servers on port 443.
- DNSRecon also logged that it got **no answer to the DNSSEC query**, which fits the "unsigned" status seen in WHOIS.

Why it matters: this single tool mapped out the DNS and mail setup. The BIND version banner is unnecessary detail, and the SPF record ends in `~all` (soft fail), which is a weaker policy than `-all` against spoofed mail.

4.2 Network Scanning with Zenmap (W2-PM5)

For the second module I installed Zenmap on my Windows PC and scanned my own lab network. The tasks were to find my local IP and subnet, list the live hosts with their IP and MAC addresses, and save the topology as a PDF.

My PC has the address [redacted], which puts it in the [redacted] subnet. I entered that range as the target and chose the **Ping scan** profile. Zenmap built the command `nmap -sn` [redacted] (Nmap 7.993). The scan ran on 10 November 2026 and covered 256 addresses in 9.81 seconds.

Result: 3 hosts were live, including my own PC.

IP address	MAC address	Notes
[redacted]	[redacted]	Up (0.010s latency). Reported by Nmap as a QEMU virtual NIC.
[redacted]	[redacted]	Up (0.0075s latency). Same MAC and vendor as [redacted]
[redacted]	[redacted]	My own PC (the scanning machine). Nmap does not print a MAC for the local host, so this is the adapter address from my PC.

After the scan I opened the **Topology** tab. It shows `localhost` in the middle with the three hosts connected around it. All three are green, which according to the Zenmap legend means "fewer than 3 open ports". Since a ping scan does not check ports at all, that colour only shows nothing was found, not that the hosts have been checked. I then used **Save Graphic** to export the topology as a PDF to my desktop.

5. Risk Analysis / Impact

From the footprinting and scanning results I noted the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	**Web technology and version details exposed**	WhatWeb reported WordPress 7.1.4, WP Download Manager 3.3.76, jQuery 3.7.1 and a public contact email (info@networkwalks.com).	Exact versions let an attacker look up known issues for that software and decide where to focus.	● **Medium**
2	**DNS server software version disclosed**	DNSRecon returned the BIND version (9.16.23-RH) for both name servers.	Version banners make it easier to match the servers against published advisories.	● **Medium**
3	**Mail and web share one IP; SPF is soft-fail; DNSSEC unsigned**	MX record points to mail.networkwalks.com on 192.232.216.135, the same address as the website. SPF ends in ~all. Whois and DNSRecon show no DNSSEC.	A shared host widens the target area, a soft-fail SPF is weaker against spoofed mail, and unsigned DNS cannot prove its answers are genuine.	● **Medium**
4	**Server IP address identifiable**	Nslookup resolved the domain to 192.232.216.135.	Shows where the web service lives on the internet.	● **Low**

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
5	**HTTP headers reveal technical detail**	Curl -I showed the /wp-json/ REST API link (with page ID 53), cache headers and the Apache server name.	Helps with technology fingerprinting and further enumeration.	● **Low**
6	**WAF technology identifiable**	Wafw00f identified ModSecurity (SpiderLabs).	Tells an attacker what protection is in place and what to try to get around.	● **Low**
7	**Domain registration details public**	WHOIS showed the registrar (GoDaddy), dates, HostGator name servers and the four client lock statuses.	Normal public data. The registrar locks are a good sign.	● **Low**
8	**Several live hosts visible on the lab LAN**	Zenmap found three live hosts on 10.0.2.0/24, two of them sharing one MAC address.	Any unexpected device on a network should be identified and explained.	● **Low**

Risk level key: ● Critical ● Medium ● Low

These are observations from information gathering, not confirmed vulnerabilities. Nothing was exploited and no vulnerability was validated in either module. Seeing a software version, an IP address or a DNS record does not mean the system is vulnerable. Further authorised testing would be needed to confirm any real weakness.

6. Recommendations

Based on what I observed, I would suggest the following.

1. **Reduce what the website tells the public**
   Review which technology and version details are visible, for example in page source, generator tags and headers, and hide what is not needed.
2. **Keep WordPress and plugins updated**
   The CMS, the Download Manager plugin and libraries such as jQuery should be updated regularly and checked against current security advisories.
3. **Review HTTP headers and the REST API**
   Remove unnecessary headers, and restrict access to /wp-json/ endpoints that do not need to be public.
4. **Hide DNS software versions**
   Ask the hosting provider (or configure the servers) so BIND does not reveal its version.
5. **Tighten the email and DNS setup**
   Consider moving SPF from `~all` to `-all` once all legitimate senders are listed, add DMARC if it is not already present, and look at enabling DNSSEC.
6. **Keep the WAF enabled and tuned**
   ModSecurity is already in place. Keep its rules current and monitor its logs.
7. **Scan internal networks regularly**
   Periodic discovery scans show which devices are active and help catch anything unexpected.
8. **Investigate unknown devices**
   Any host that cannot be explained should be identified and verified, including hosts that share a MAC address.

9. **Document the network**
   Keep an up-to-date list and topology diagram of devices, IPs and MAC addresses.
10. **Only test with authorisation**
    Reconnaissance and scanning should be done only on systems where permission has been given.

7. Conclusion

In Week 2 of the program I worked through footprinting and network scanning. On networkwalks.com, six tools were enough to learn the registrar and hosting provider, the WordPress and plugin versions, the IP address, the firewall in front of the site, and the layout of the DNS and mail records. None of this needed anything more than public queries.

With Zenmap I found my local address and subnet, discovered three live hosts on [redacted], recorded their IP and MAC addresses and saved the topology as a PDF. The matching MAC addresses on two hosts were an interesting detail and a good reminder that scan results need interpreting, not just copying.

The main thing I take from these modules is how much can be learned before any attack is attempted, and why defenders should look at their own systems the same way. I also learned that a good report explains what was done, what was found, what it means and what to do about it, and that all of this work must stay inside an authorised scope. These activities were done as part of an assigned educational lab.

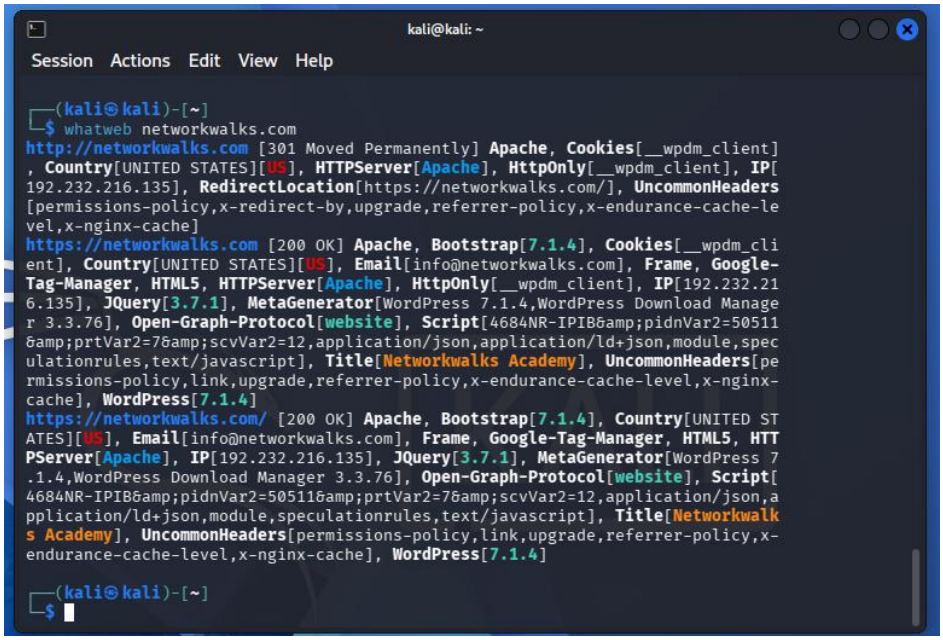
8. Evidences Collected

Terminal output is included as a screenshot exactly as it appeared in Kali. The Zenmap screenshots follow.

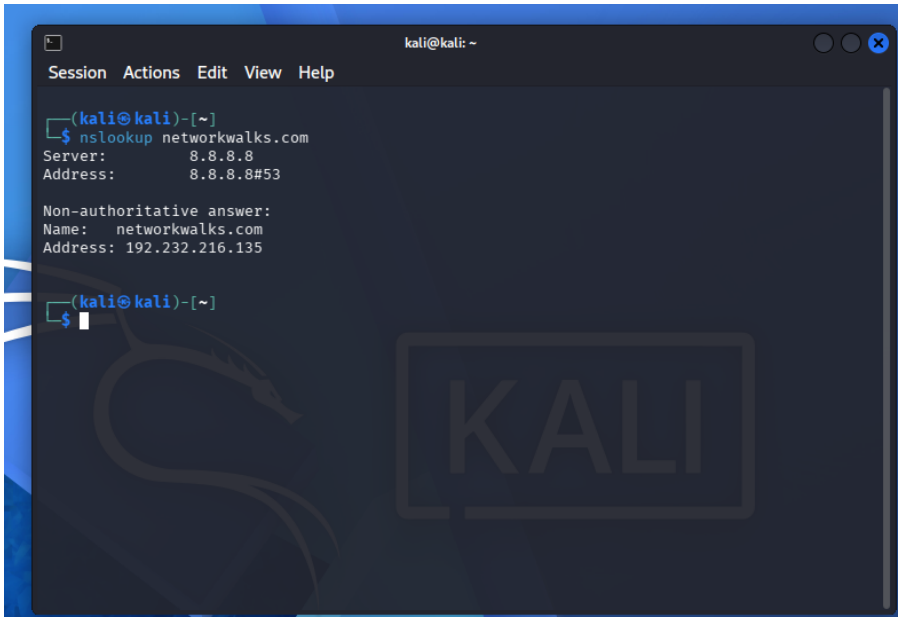
8.1 WHOIS: whois networkwalks.com



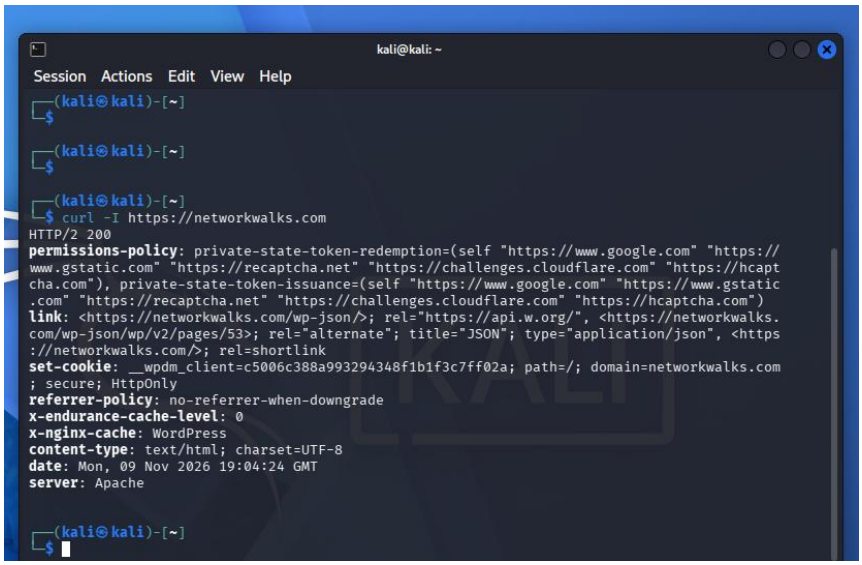
8.2 WhatWeb: whatweb networkwalks.com



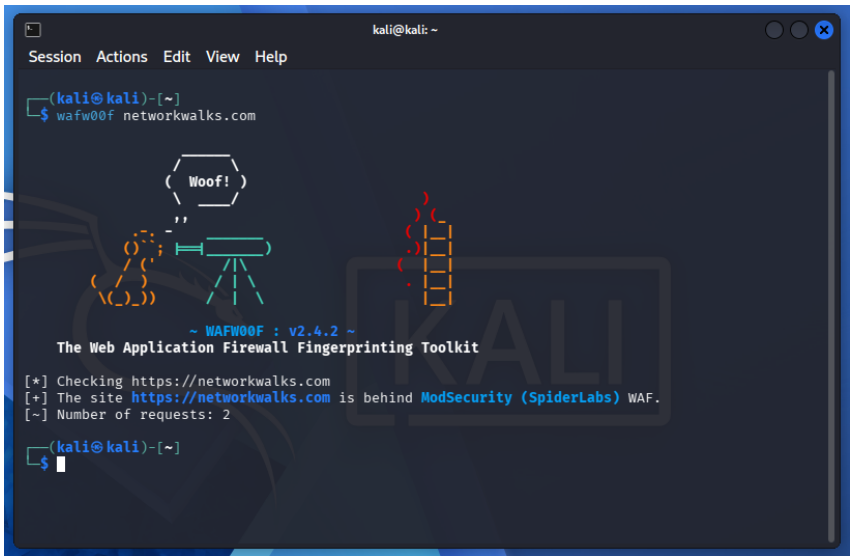
8.3 Nslookup: nslookup networkwalks.com



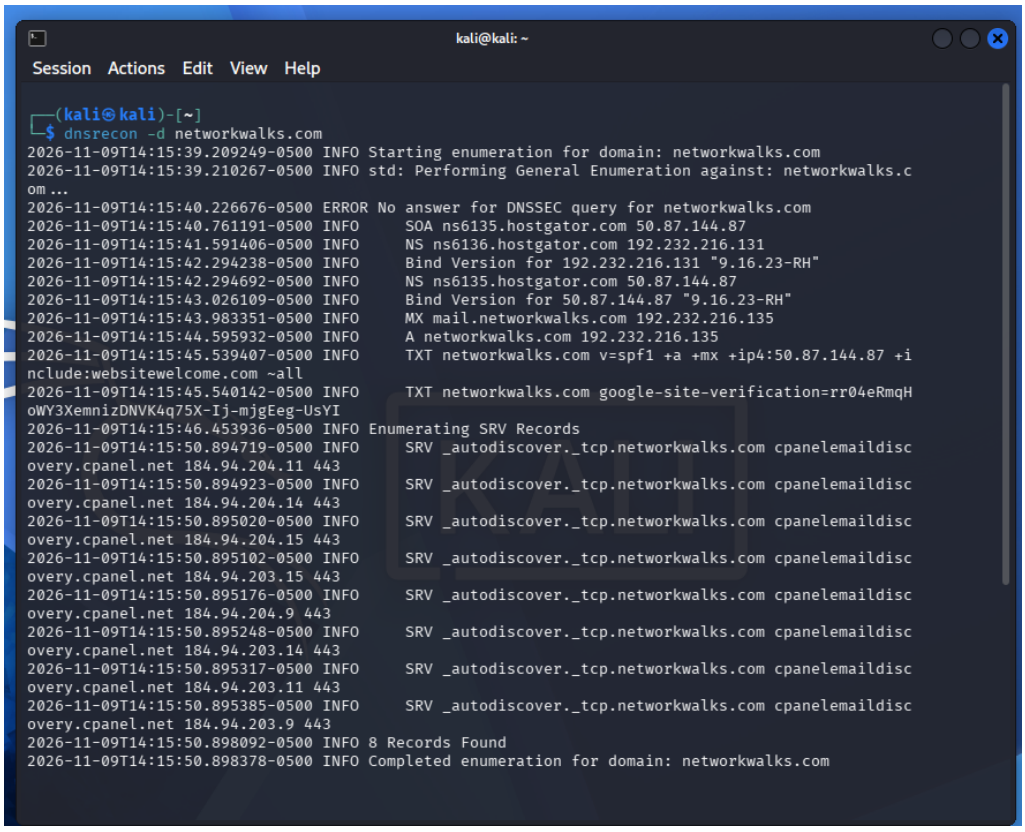
8.4 Curl: curl -I https://networkwalks.com



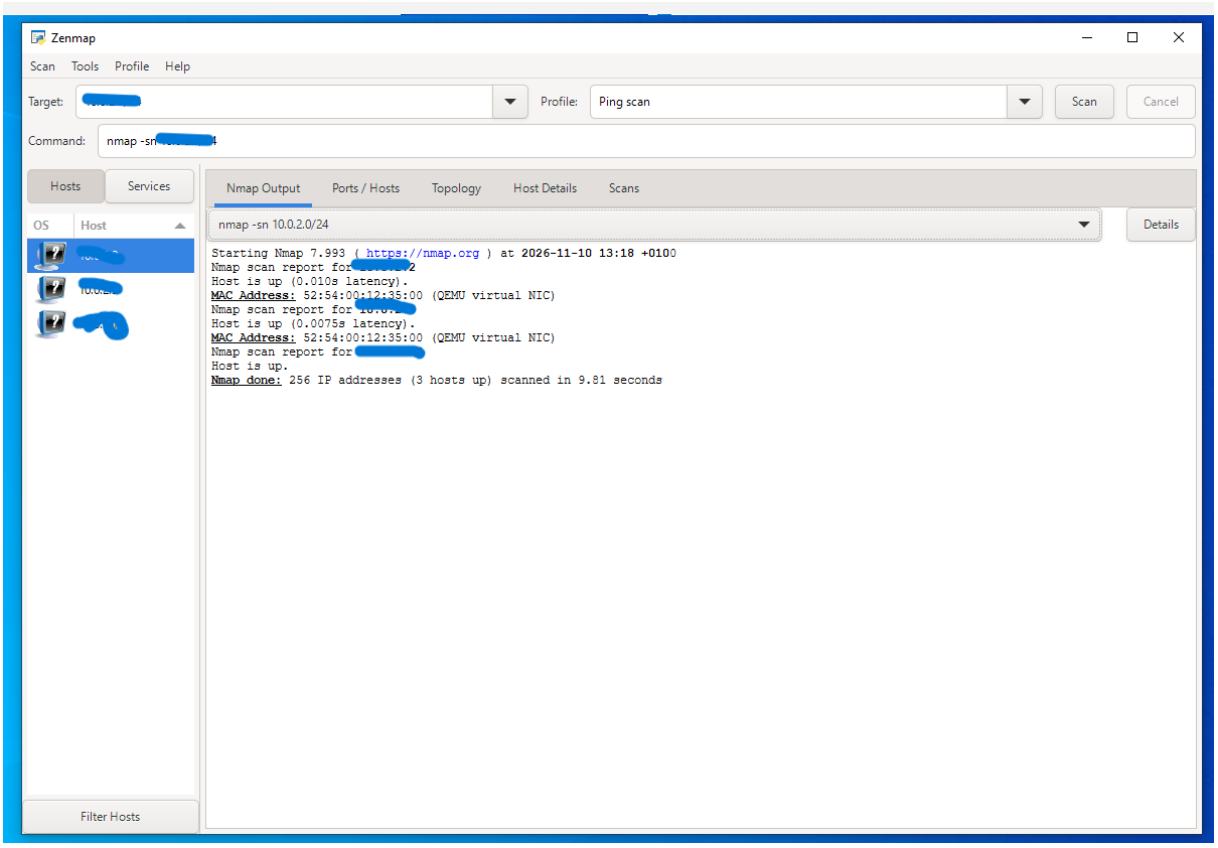
8.5 Wafw00f: wafw00f networkwalks.com



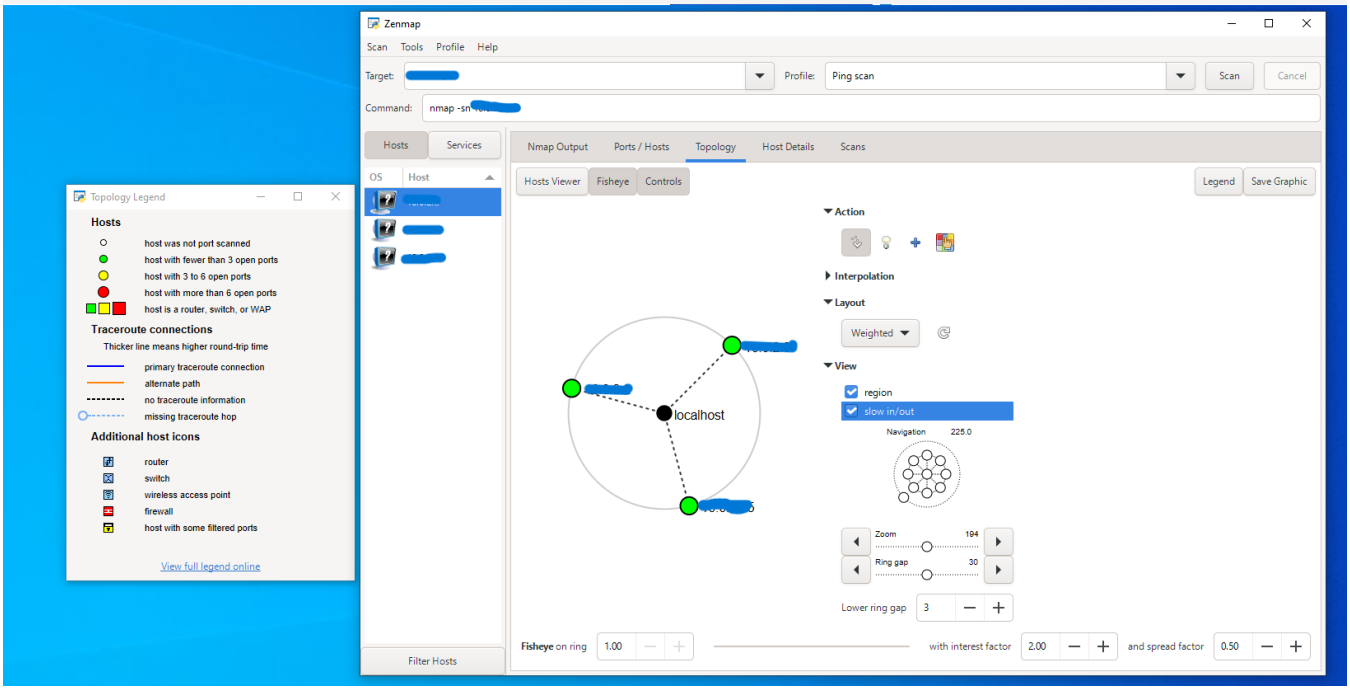
8.6 DNSRecon: dnsrecon -d networkwalks.com



8.7 Zenmap ping scan: nmap -sn 10.0.2.0/24



8.8 Zenmap topology view with legend



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Project Information

Program Name: Cybersecurity program at Networkwalks | **Week:** 02